

Runguo Li

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Research Interests

Machine learning systems for large models: efficient LLM inference (KV-cache compression and management, MoE serving), distributed training and RL post-training infrastructure, and the correctness of ML compilers and GPU kernels. More broadly: how systems constraints interact with model behavior, such as how memory budgets shape long chain-of-thought reasoning, informed by prior work on LLM reasoning, multimodal learning, and agent safety.

Education

University of Illinois Urbana-Champaign (UIUC)

M.S. in Information Science (expected Jul 2028)

Aug 2026 – Present

Urbana-Champaign, IL, USA

Shanghai University of Finance and Economics (SUFU)

B.S. in Data Analytics, GPA: 3.78/4.0

Sep 2022 – Jun 2026

Shanghai, China

Publications and Preprints

(* equal contribution)

- [1] **Runguo Li. Spend Bytes on Breadth: Precision–Count Trade-offs for Decode-Time KV Compression in Long Chain-of-Thought Reasoning.** *arXiv preprint arXiv:2610.05685, 2026.* [[arXiv](#)] [[PDF](#)] [[Code](#)]
Single-author. Splits a fixed KV byte budget between token count and precision (BreadthKV); beats eviction alone in 17 of 18 settings across three reasoning models and four benchmarks.
- [2] Zhi Yang*, **Runguo Li***, Qiqi Qiang, Jiashun Wang, Fangqi Lou, Mengping Li, Dongpo Cheng, et al. **FinVault: Benchmarking Financial Agent Safety in Execution-Grounded Environments.** *arXiv preprint arXiv:2601.07853, 2026.* [[arXiv](#)] [[Code](#)]
First execution-grounded safety benchmark for financial agents (31 sandbox scenarios, 963 test cases); attack success rates reach 50%.
- [3] Yeheng Chen*, **Runguo Li***, et al. **VeriBRT: Plan-Guided, Evidence-Based Automated Bug Reproduction.** *Under review, 2026.* [[PDF](#)] [[Code](#)]
Plan-guided bug-reproducing test generation; reproduces 341 of 433 issues on SWT-Bench Verified, 9.7 points above the strongest same-model baseline.
- [4] Yutao Hou*, **Runguo Li***, et al. **Reasoning-Visual Critical Token Fine-Tuning for Multimodal Reasoning.** *Under review, 2026.* [[PDF](#)] [[Code](#)]
Selects which chain-of-thought tokens receive supervision using reasoning-relevance and visual-sensitivity signals (RVCFT).

Research Experience

University of Illinois Urbana-Champaign

Research Intern, advised by Prof. Minjia Zhang

Aug 2026 – Present

Urbana-Champaign, IL, USA

- Efficient ML systems for large models: decode-time KV-cache compression for long chain-of-thought reasoning (single-author preprint [1]), model compression and sparsity, and kernel- and hardware-aware optimization.
- Upstream systems work grounded in this research: reproducing failures in production inference and training stacks and contributing fixes with regression tests (see Open-Source Contributions).

Shanghai Jiao Tong University

Research Intern, advised by Prof. Xiaodong Gu

Mar 2026 – Jul 2026

Shanghai, China

- LLMs for code: repository-level program understanding, automated bug reproduction, and coding-agent planning.
- Co-first author of **VeriBRT** [3], which represents issue-report behavior as a persistent Two-Axis Plan and uses reliability-aware evidence for bug-reproducing test generation; built the SWT-Bench Verified evaluation harness.

Tencent Youtu AI Lab

Research Intern

Oct 2025 – May 2026

Shanghai, China

- Co-first author of **RVCFT** [4], a selective-supervision method for multimodal chain-of-thought fine-tuning.
- LLM-driven adversarial data synthesis and low-resource, multi-label content-safety evaluation; contributed to a unified multilingual moderation model (XLM-RoBERTa backbone).

Open-Source Contributions

Merged pull requests to ML inference, training, and compiler systems (project names link to the PR lists):

- **vLLM**: fixed `int4_per_token_head` KV-cache quantization for non-power-of-two head sizes, where the Hadamard-rotation write aborted the engine core (#56198).
- **SGLang**: deferred DeepEP import-time failures to first use, so host misconfigurations no longer crash every server process, including dense models that never use DeepEP (#40671).
- **DeepSpeed**: fixed a ZeRO-3 rollout deadlock from unsynchronized generation stopping; blocked partial Hybrid Engine policy injection for unsupported architectures (validated on H200 and MI250).
- **Triton**: stopped nested-loop fusion from trusting an out-of-loop `llvm.assume` that removed a zero-trip guard and could cause incorrect memory writes.
- **FlashAttention**: fit the SM90 CuTe forward tile to the block-sparse block size; skipped the causal mask on unmasked KV blocks (580 $\rightarrow$ 477 inner-loop instructions on H200).
- **colibri**: multi-drive expert streaming for a 510 GB MoE checkpoint, physical-core OpenMP sizing (18.7 $\times$ speedup in four engines), and a fix for 6.25 GiB of unbudgeted KV cache.
- **JAX**: made `jax.scipy.linalg.lu` support documented batched inputs (#40931).
- **Also**: [verl](#) (FSDP critic loading), [LMCache](#) (allocator free-list corruption), [vLLM-Omni](#) (Qwen2.5-Omni engine crash), and [Apache TVM](#) (ONNX Hardmax with symbolic shapes).

Honors and Awards

- National Encouragement Scholarship | People's Scholarship
- Shanghai Municipal Scholarship | Social Practice Scholarship

Skills

- **Programming**: Python, C, CUDA, Triton, SQL, Bash.
- **Systems and frameworks**: PyTorch, JAX, vLLM, SGLang, DeepSpeed, Megatron-LM, FSDP, verl, MLIR; GPU profiling on NVIDIA H200 and AMD MI250.